CREDIT ASSISTANT

AI-DRIVEN CREDIT INTELLIGENCE & HEALTH PLATFORM

DOCUMENT: Technical Specification & Architecture
REGULATORY FOCUS: Indian Banking (RBI, CIBIL, Experian)
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EXECUTIVE ABSTRACT: Credit Assistant is an enterprise-grade financial intelligence platform engineered specifically for the Indian credit ecosystem. Regulated under guidelines from the Reserve Bank of India (RBI) and scored under bureaus (TransUnion CIBIL, Experian India), the platform unites deterministic algorithmic modeling, low-latency Groq AI inference, real-time What-If score simulation, and dynamic ReportLab PDF audit compilation to deliver institutional-grade credit diagnostics and actionable recovery roadmaps for retail borrowers.

1. Systemic Problem Statement in the Indian Retail Credit Landscape

In India, access to prime personal, auto, and home financing is governed strictly by the 300–900 bureau scoring range. Despite rapid credit card penetration and retail lending growth, consumers face acute systematic hurdles:

Challenge	Systemic Root Cause in Indian Banking	Borrower Financial Consequence
Opaque Bureau Scoring	Bureaus (CIBIL/Experian) do not reveal live point deductions for utilization spikes or DPDs.	Borrowers blindly trigger score drops by crossing the 30% credit card utilization threshold.
Unmonitored FOIR / DTI	Indian banks enforce strict Fixed Obligation to Income Ratios (FOIR capped at 40-50%).	Uninformed loan applicants face outright rejections, generating hard bureau inquiry penalties.
Compounding 42% APR Debt	Revolving credit card interest in India charges 3.5% to 3.75% per month (42% to 45% APR).	Paying only minimum dues leads to exponential compounding balance traps and debt spiraling.
Western Advice Inapplicability	Generic financial tools ignore Indian specifics: e-NACH auto-debits, CIR disputes, and 30/60 DPDs.	Borrowers fail to utilize formal Indian statutory grievance dispute channels.

2. Strategic Engineering Objectives & Target Metrics

Target Objective	System Implementation Architecture	Target Metric / KPI
Automated Risk Modeling	Deterministic FOIR & CUR algorithmic engine with banking 5% card rule.	Zero calculation discrepancy across all income bands.
Pre-Emptive Simulation	Reactive What-If simulation slider matrix calculating instant score projections.	< 15ms simulation execution time.
AI-Driven Advisory	Groq Cloud API integration running Qwen 3.8 / LLaMA structured prompt engine.	< 1.5s turnaround for full 5-step recovery roadmap.
Statutory Audit PDF	Vectorized ReportLab document generation engine with compliance notices.	Downloadable multi-page institutional audit in < 500ms.

3. End-to-End System Architecture & Infrastructure Design

The platform adopts a decoupled client-server architecture designed for high availability, deterministic calculations, and sub-second AI inference. The React client communicates with the asynchronous FastAPI backend via JSON REST APIs, utilizing SQLite for lightweight local verification and full PostgreSQL dialect compatibility for cloud deployments.

Tier	Technology Component	Architectural Responsibility & Implementation Details
Frontend Presentation	React 18 + Vite SPA Lucide Icons + Canvas Confetti	Client-side dashboard state management, real-time What-If sliders, SVG gauge visualization, dynamic AI chat drawer, and milestone celebration triggers.
API Application Core	FastAPI (Python 3.10+) Uvicorn ASGI Server	High-throughput non-blocking request pipeline, automated OpenAPI/Swagger documentation, CORS security middleware, and dual `/health` and `/api/health` diagnostic probes.
Financial Modeling Engine	Algorithmic Math Engine (`calculator.py`)	Deterministic calculation of FOIR/DTI ratios, credit card utilization, tier classification, consecutive on-time payment boosts, and avoided compounding interest.
Artificial Intelligence	Groq Cloud Inference Engine Qwen 3.8 / LLaMA Models	High-speed token generation for executive bottleneck diagnosis, 5-step recovery roadmaps, and conversational borrower chat advisory with automatic heuristic fallbacks.
Document Generation	ReportLab 5.0+ Engine (`pdf_service.py`)	Server-side streaming compilation of vector-grade official multi-page Credit Health Audit PDF documents with custom color scales and statutory RBI compliance disclaimers.
Persistence & ORM	SQLAlchemy 2.0 ORM SQLite / PostgreSQL	Entity relationship management, transactional session pools, foreign key cascade constraints, and chronological credit progression checkpoint logging.

4. Relational Database Schema & Data Models

The relational data model enforces strict referential integrity across borrower identities, snapshot profiles, historical trend logs, and cached AI consultation outputs:

Entity Table	Primary & Foreign Keys	Key Attributes & Data Types	Relationship Cardinality
users	PK: id (Integer)	name (String), email (Unique String), phone (String), created_at (DateTime)	1 : 1 with financial_profiles 1 : N with history & AI logs
financial_profiles	PK: id FK: user_id (Unique)	credit_score (Int), monthly_income (Float), monthly_expenses (Float), credit_card_limit (Float), credit_card_balance (Float), total_loan_emi (Float), active_loans_count (Int), missed_payments_24m (Int), dti_ratio (Float)	1 : 1 parent-child constraint with automatic cascade deletion
credit_history_log	PK: id FK: user_id	month_label (String), credit_score (Int), utilization_ratio (Float), dti_ratio (Float), total_debt (Float), note (String), recorded_at (DateTime)	1 : N chronological checkpoints for trajectory visualizations
ai_analysis_logs	PK: id FK: user_id	executive_summary (Text), bottlenecks_json (Text), roadmap_json (Text), raw_response (Text), created_at (DateTime)	1 : N consultation history log for audit and re-read

5. RESTful API Contract & Endpoints Specification

The FastAPI backend exposes standard REST endpoints adhering to JSON data contracts and HTTP semantics:

Verb	Endpoint Route	Input / Payload	Output Response & Status
GET	/health, /api/health	None (Diagnostic ping)	Status 200: DB latency ms, uptime, environment
GET	/api/users	None	Status 200: Array of all registered borrowers
POST	/api/users	JSON: Borrower profile payload	Status 201: Created user with initialized profile & ratios
PUT	/api/users/{id}/profile	JSON: Updated income/debt fields	Status 200: Recalculated FOIR, utilization & tier
POST	/api/users/{id}/simulate	JSON: Simulation delta sliders	Status 200: Projected score, interest saved, deltas
POST	/api/ai/consult/{id}	None (Context derived from DB)	Status 200: Executive diagnosis & 5-step JSON roadmap
POST	/api/ai/chat/{id}	JSON: User query message	Status 200: Conversational advisory response
GET	/api/report/pdf/{id}	None	Status 200: Binary application/pdf document stream

6. Algorithmic Financial Engine & Underwriting Mathematics

The algorithmic calculation engine (`backend/calculator.py`) evaluates borrower debt burdens using strict formulas codified from Reserve Bank of India (RBI) prudential guidelines and Indian Credit Information Companies (CIBIL/Experian):

A. Fixed Obligation to Income Ratio (FOIR / DTI) — Indian 5% Card Rule

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Mathematical Formula:
    Total Monthly Obligations (Rs.) = Total Loan EMIs + (Credit Card Balance × 0.05)
    FOIR (%) = [ Total Monthly Obligations / Monthly Income ] × 100
```

Underwriting Norms: ≤ 35% is Prime / Optimal; 35%–50% is Moderate / Strained; > 50% High Risk (Loan Rejection Tier).

B. Credit Utilization Ratio (CUR) — 30% CIBIL Threshold

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Mathematical Formula:
    CUR (%) = min( 100, [ Credit Card Balance / Credit Card Limit ] × 100 )
```

CIBIL Golden Rule: CUR ≤ 30% generates optimal score growth; CUR > 50% triggers steep point deductions.

C. Indian Bureau Tier Matrix & Lending Implications

Score Range	Bureau Classification	Underwriting Outcome & Lending Terms in India
750 – 900	**Excellent (Prime)**	Pre-approved home & auto loans, lowest interest rate spreads (Repo + 2.05%), 100% processing waiver.
700 – 749	**Good**	Standard retail credit approvals; regular documentation and standard processing fees.
650 – 699	**Fair (Subprime)**	Higher risk-premium spreads (+150-250 bps); mandatory collateral or co-borrower requirement.
300 – 649	**Poor (Distressed)**	Outright rejection for unsecured lending; eligibility limited strictly to FD-backed secured credit cards.

7. Dynamic What-If Credit Score Simulation Formulation

The real-time simulator computes score trajectory adjustments by applying weighted deltas derived from bureau algorithms:

Simulation Factor	Mathematical Formulation & Algorithmic Rule	Maximum Impact
Utilization Recovery	Delta = (Current CUR - Simulated CUR) × 0.40. If CUR drops from >50% to ≤30%, an additional +15 point milestone bonus is awarded.	**+45 Points**
FOIR Reduction	Delta = (Current FOIR - Simulated FOIR) × 0.35. If FOIR decreases by >5%, an extra +8 point debt relief bonus is granted.	**+30 Points**
Seasoning Bonus	Delta = Consecutive On-Time Payment Cycles × 4 points per verified reporting cycle.	**+35 Points**
High-APR Savings	Avoided Compounding Interest = Repaid Principal × (0.42 / 12) × 6 months. Highlights cash savings against standard 42% annualized card revolving fees.	**Direct Liquidity Relief (Rs.)**

8. Groq AI Advisory Engine & Consultation Architecture

The AI Consultation Engine (`backend/groq_service.py`) utilizes Groq Cloud LPU acceleration to execute contextual credit diagnostics in sub-second latency, formulating a personalized, sequential 5-step recovery roadmap:

Roadmap Step	Target Area	Actionable Strategy Codified for Indian Banking Realities
Step 1	**CIBIL CIR Dispute Resolution**	Initiate an online Credit Information Report (CIR) dispute via the CIBIL portal for erroneous DPDs, duplicated loan accounts, or unsettled status marks wrongly reported by lending institutions.
Step 2	**Utilization Optimization (≤30%)**	Pay down revolving credit card balances below 30% before the billing cycle statement date. Request sanctioned credit limit upgrades to expand denominator headroom without spending increases.
Step 3	**e-NACH Auto-Debit Mandates**	Configure electronic National Automated Clearing House (e-NACH) auto-debit mandates across salary accounts to eliminate human oversight and permanently avoid 30/60 DPD (Days Past Due) bureau marks.
Step 4	**Debt Avalanche Strategy**	Target aggressive debt elimination towards high-APR revolving credit cards (compounding at ~42% APR) while maintaining minimum contractual obligations on low-cost secured loans (home/car loans at 8-9%).
Step 5	**Credit Mix & Vintage Seasoning**	Maintain oldest active credit cards to protect credit vintage. Balance portfolio by pairing unsecured debt with secured assets (e.g. gold loan, auto loan) to achieve an optimal 60:40 secured-to-unsecured ratio.

9. Real-Time Conversational AI Advisor (AIChatDrawer)

Borrowers can query the AI Advisor dynamically via a slide-out drawer interface. The backend injects verified profile data (CIBIL score, FOIR, missed payments, active loans) as system context, ensuring answers reference the borrower's exact numbers rather than generic platitudes. Complete session dialogues are stored in memory and summarized for ongoing continuity.

10. Automated End-to-End Test Suite & Verification Results

The system incorporates an end-to-end integration test suite (`test_e2e.py`) validating all 8 core API services and data workflows across simulated borrower profiles:

Test Case ID	Verification Scenario	Validation Criteria	Status
TC-001	Diagnostic Health Probe	Status 200, DB latency < 50ms, uptime verification	**PASS**
TC-002	Borrower Onboarding & Baseline	Profile created, FOIR and CUR accurately computed	**PASS**
TC-003	Profile Modification & Recalc	Financial values updated, derived ratios recomputed	**PASS**
TC-004	Chronological History Fetch	Array of historical logs returned with timestamps	**PASS**
TC-005	What-If Simulator Engine	Score delta accurately projected with compound savings	**PASS**
TC-006	ReportLab PDF Compilation	Binary PDF byte-stream verified (> 5 KB, valid PDF header)	**PASS**
TC-007	Groq AI Consultation Flow	Executive summary & 5-step structured roadmap parsed	**PASS**
TC-008	Conversational Chat Advisory	Interactive response returned with contextual memory	**PASS**

11. Production Deployment Topology & Execution Setup

Environment	Host & Configuration	Execution Commands / Orchestration
Backend API	Render / AWS ECS Python 3.10+, Uvicorn ASGI	py -m uvicorn backend.main:app --host 0.0.0.0 --port 8000 Render web service managed via render.yaml.
Frontend SPA	Vercel Edge Network React 18 + Vite Production Bundle	npm run build -> static export to dist/. Automatic proxy routing configured via vercel.json.
Database	SQLite (Local) / Supabase Postgres	Resolved automatically via DATABASE_URL environment variable with seamless ORM migration.

12. Future Innovation Roadmap & Strategic Milestones

Phase	Strategic Initiative	Technological Enhancement & Market Impact
Phase 1 (Q1)	**Account Aggregator (AA) Integration**	Connect with RBI-licensed Account Aggregators (Setu, Anumati) to ingest real-time bank statement cash flows automatically.
Phase 2 (Q2)	**Vernacular Multi-Lingual Advisory**	Deploy Hindi, Marathi, and Tamil conversational agents via Groq to democratize credit intelligence for tier-2/3 Indian cities.
Phase 3 (Q3)	**Automated CIR Dispute Filing Bot**	Enable one-click submission of pre-drafted grievance letters directly to TransUnion CIBIL and banking ombudsmen.

ENGINEERING CERTIFICATION: This document certifies the technical architecture, mathematical modeling, and operational verification of the Credit Assistant Platform. All formulas strictly conform to Indian commercial banking standards (RBI Master Directions / CIBIL / Experian). Validated by automated end-to-end integration tests TC-001 through TC-008.